Design of high fidelity combinatorial libraries

October 25, 2026

Abstract

Abstract — yours. The numbers that are settled and quotable, if you want them to hand: library pass 53.5 % against 30.5 % sampled over 9 backbones; the chimera objective at 93.4 % on 3 of 3; 3135 wells assayed at a 67 % register rate, median 29 mutations from the parent; and the cost identity, $S = (B/3Lc)^k$.

Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	drafted
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	drafted
`Makefile`	`make pdf`; SVG→PDF via Inkscape	drafted
`refs.bib`	Bibliography; all entries machine-retrieved	drafted
Front matter		
`abstract.tex`	∼125-word abstract, written last	scaffold
`intro.tex`	Four-paragraph introduction arc	scaffold
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	drafted
Results — the spine		
`results/frontier.tex`	Move 1: RL moves the diversity/fidelity frontier. Public benchmark, referee-reproducible	scaffold
`results/fragments.tex`	Move 2: split + recombine → combinatorial library. Computational half measured; physical half in `experimental.tex`	scaffold
`results/experimental.tex`	Move 3: that library ordered and assayed. 3,135 recombined PETase designs; zero-shot AF3 vs measured activity, and its failure to transfer to a second serine hydrolase	drafted
Methods		
`methods/method.tex`	§3, the fragment-RL objective (your draft)	scaffold
`methods/overview.tex`	Detailed pipeline; partly duplicates §3	scaffold
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	scaffold
Not in this build		
`notebook.tex`	Separate document: the E1–E21 record	drafted
`experiments.tex`	Arms, selection rules, slate E1–E21, algorithm audit; **\input by notebook.tex, not by main.tex**	drafted
`campaigns.tex`	Separate document: one panel per campaign, the diversity/pass Pareto, the cost model, and three candidate narrative arcs	drafted
`library_notebook.tex`	Separate document: the library arc, L1–L5	drafted
`library_experiments.tex`	Fragment retention, recombination, granularity, the transfer gap; **\input by library_notebook.tex**	drafted
Archived — see `archive/README.md`		
`archive/*`	The heme and protease arcs, one superseded outline, and the *plan* for a three-round closed-loop PETase campaign that has no data. The PETase data that does exist is in `results/experimental.tex`	drafted

The paper and the record are two documents. `main.tex` builds the paper (`make pdf`); `notebook.tex` builds the experimental record (`make notebook`); `make all` builds both. The split is deliberate: the notebook keeps pre-registrations next to the results that falsified them, which is what makes a retraction checkable and what a paper cannot carry and stay readable. A claim moves from the notebook to the paper by being rewritten as prose, never by \input.

Two companion files live in `paper/` but are deliberately *not* part of either build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

As the age of de novo design begins to mature, we have access to a multitude of tools which allow us to design enzymes. The sequence to function landscape surrounding a given seed sequence is not well understood. There are clear pathways and routes of evolution that exist to evolved sequences along a specified trajectory. Inverse folding models allow us to sample from the space of possible sequences for a given backbone [1–3]. This is useful as small changes in sequence space can amount to orders of magnitude change in function. For a given function we struggle to understand why certain sequences for a given backbone are functional while others are not. Filtering by looking at the geometry of the active site from a structure prediction model like AF3 [4] can have some predictive power for certain functions. Using inverse folding models to propose many sequences, it is possible to accumulate millions of potential candidate sequences which pass all of the filtering metrics, but nearly impossible to rank which of those will be most functional in an assay readout. Furthermore the cost of ordering a library with millions of unique sequences is cost prohibitive.

In this work we will propose a method for designing large scale high fidelity combinatorial libraries which is cost effective and feasible to generate.

> Two things this section can draw on when you come back to it. (i) "nearly impossible to rank" is *measured* in this project, not assumed: on PETase the best zero-shot AF3 metric reaches $\rho = +0.60$, and every one of the ten strongest predictors inverts on a second serine hydrolase (ρ-of-ρs $= -0.604$, 10 of 49 concepts agreeing in sign). That is the strongest available support for the sentence and it is currently unused here. (ii) "cost prohibitive" has a number: at Twist eBlock pricing a 200-aa enzyme is \$42 per full-length gene, so a million unique sequences is \$42M, while the same million as a $k = 6$ combinatorial library is under \$500. See `campaigns.pdf` §11.

2 Prior Work

Library design has often consisted of error prone PCR to sample various amino acids at different positions. This can be done in a targeted way or more generally over the entire scaffold. Many directed evolution campaigns often use error prone PCR to build a large library. Many mutations are often deleterious and modern tools like structure prediction allow us to filter out many of these obviously harmful mutations. In addition gene fragment shuffling is another way to go about making diverse libraries. Methods like SCHEMA from Frances Arnold's group aimed to find the best points at which to split enzymes for combinatorial shuffling [5, 6]. More modern work like that from Sarel Fleishman [7] and SAMPLE from Phil Romero [8] demonstrate how we can use combinatorial libraries to do active learning style iteration. As autonomous labs are being developed around the globe and we are preparing to scale data generation for building better protein function predictors. We have a need for generating high fidelity large scale libraries for evaluation. Here we will present an approach for post-training ProteinMPNN [1] to propose protein fragments which combine nicely with one another to create a high fidelity and highly diverse library.

> Left exactly as written, including the two places you may want another pass: "As autonomous labs . . . predictors." is a subordinate clause with no main verb and currently reads as a fragment, and the following sentence may be its intended completion. Also: the AME experiments in this paper use LigandMPNN, not ProteinMPNN — the PETase work uses ProteinMPNN. If the claim is about the method rather than one checkpoint, "an MPNN sequence designer" covers both; if it is about what was run, the two need naming separately.

3 Method

We find that we can train for the objective of maximizing the ability of fragments from ProteinMPNN[**TODO: sentence is incomplete — maximizing their ability to *what*? recombine? fold when recombined?**]. To do this we take a protein of length L and we split it up into K fragments such that each fragment f_i is approximately L/K amino acids long. Fragments are purposely built to be equal length to make DNA assembly easier, and empirically we do not find that it matters so much how you split up the protein (i.e. secondary structure, domains, etc.). We find that it is possible to find a space of protein fragments that are highly combinable while also still diverse. By treating the ProteinMPNN decoder as a policy conditioned on the encoded backbone, we can do a short RL trajectory to align the decoder to the space of highly compatible combinatorial fragments. For an encoded backbone we decode B sequences, then we split each into fragments, sample the fragments into a larger pool of sequences $B \times 4$ such that each fragment is seen 4 times in different combinations. When each sequence is folded a score is computed by measuring RMSDs in the active site and confidence. We aggregate these into a score for each fragment by averaging the score across all the sequences that fragment was contained in. We provide rewards at the fragment level when we go to update the policy. This means that each fragment in a given sequence will receive a different reward. In practice there might be some issues here because as the fragments are being decoded they will condition on other fragments as well, but empirically this objective seems to be stable and optimizes with the GRPO objective [9].

Three details from the implementation that this paragraph does not yet state, all of which a referee will ask for. (i) **The pool contains unrecombined parents.** `build_chimeras` maps (ρ, i) to parent $(i + \rho j) \bmod B$, so the $\rho = 0$ rotation returns the parent itself — 32 of every 128 chimeras at every k. Whether that is deliberate (an anchor on the parental distribution) is worth one sentence either way. (ii) **Only $k = 4$ was ever trained.** Everything in this paper about granularity re-cuts $k = 4$ designs post hoc. (iii) **The fold is best-of-5.** That is the AME benchmark's own unit; a single-sample run measures something the benchmark does not.

4 You cannot rank what you can make

Section is yours to frame. What is below is the evidence, already written. What it is being asked to do is new: it has to carry the intro's "nearly impossible to rank which of those will be most functional" from assertion to measurement, and hand off to the cost section (Section 5) — if you *could* rank them you would not need a library, and if ranking fails the library is the only remaining move.

The figure that was missing now exists. The cross-enzyme comparison had no figure script and was an inline table; it is drawn as Figure 1 below, from `paper/figures/campaign_panels.py`.

Zero-shot prediction is lightly predictive

Every design was folded by AF3 in two states — the Michaelis complex and a covalent *tetrahedral adduct* — and scored on 52 confidence and geometry metrics. The adduct is not an acyl-enzyme intermediate, although the column is named `acyl_*`: AF3 strips the leaving hydrogens but keeps the leaving oxygens, so the acid –OH survives and the bonded carbon stays 4-coordinate. Both enzymes in the transfer comparison below model the same species, so that comparison is unaffected — but the state is the adduct, and calling it the acyl-enzyme intermediate would misstate which chemistry the predictor was scored on. Table 1 gives the ranking against measured activity.

Three readings, in the order that matters.

The useful signal is local and state-specific, not global. pTM and ipTM — the metrics a

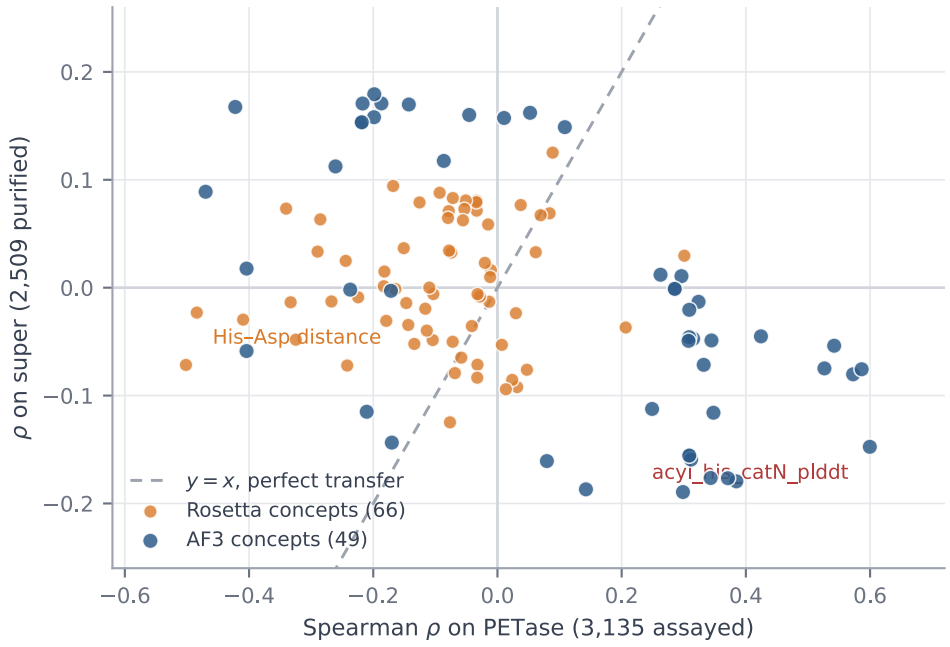


Figure 1: Each point is one metric, plotted by how well it predicted activity on PETase against how well it predicted on a second de novo serine hydrolase. On the diagonal a metric transfers; in the off-diagonal quadrants it points the wrong way. The 49 shared AF3 concepts anti-correlate at ρ-of-ρs $= -0.604$, with 10 agreeing in sign against $\sim$24 expected by chance. The 66 Rosetta concepts fail more mildly ($+0.035$, 33 of 66): no transferable signal rather than an inverted one.

design pipeline reaches for by default — reach $\rho \approx 0.35$. Per-atom pLDDT at the catalytic histidine of the *tetrahedral adduct* reaches 0.60. The gain comes from scoring the chemically relevant state at the chemically relevant atoms, which is an argument for building the covalent intermediate into the oracle rather than for folding harder.

It is a real signal, and it is light. Selecting the top 5% of the library by active-site pLDDT returns a top-decile-active design 28% of the time against a 10% base rate: a 2.8× enrichment, falling to 1.9× if 25% of the library is kept. That is worth having when the alternative is random ordering, and it is nowhere near a filter one would trust to replace an assay.

And a cheaper predictor beats it. Mutation count from the parent, which requires no structure prediction at all, ranks activity at $\rho = -0.65$ — better than any AF3 metric in the table (Figure 2c). This is the comparison a zero-shot claim has to clear and it does not clear it. What survives is that the AF3 signal is not merely that confound re-expressed: restricted to the 2,539 designs at least 20 mutations from the parent, active-site pLDDT still gives $\rho = +0.59$. Within single 5-mutation bands it ranges from $+0.05$ nearest the parent to $+0.59$ at 30 to 34 mutations, and is indistinguishable from zero in the three most distant bands (-0.01 to $+0.09$ beyond 50). Prediction adds something to distance; it does not dominate it — but it adds it only in the middle of the range, and the band-wise view is what shows that the pooled $+0.59$ is not uniform.

. . . and it does not transfer to a second enzyme

The $\rho = 0.60$ above is a property of this scaffold, not of the method. The same two-state fold and the same metric pipeline were run against a second, unrelated serine hydrolase — `super`, a 160-aa de

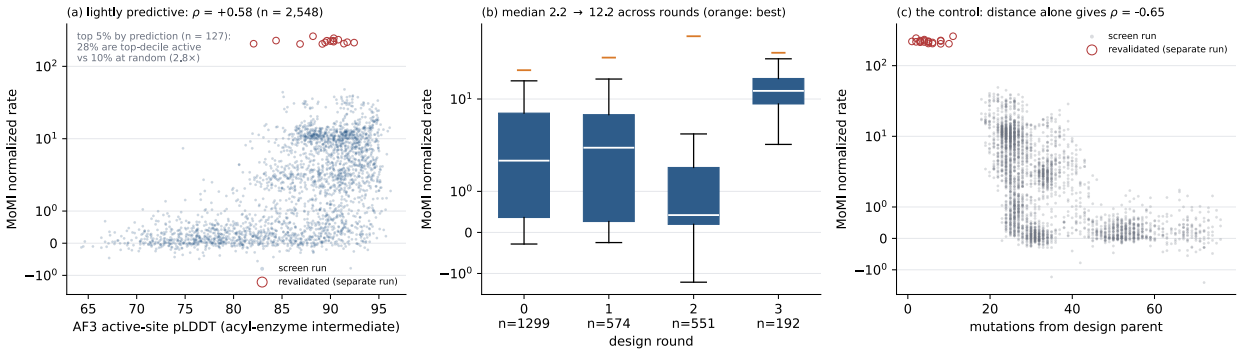


Figure 2: **Structure-prediction confidence against measured activity for 3,135 recombined PETase designs.** (a) The strongest zero-shot signal is local and state-specific: per-atom pLDDT over the active site of the *tetrahedral adduct*. Selecting the top 5% by this metric returns top-decile actives at 2.8× the base rate. (b) Activity by design round, on the single screen run. (c) The control: distance from the design parent ranks activity better than any prediction does. The 20 open red points are a selected set re-assayed in a different run of the same assay and are never pooled into a correlation.

novo design (Ser128/His95/Asp53, 4-methylumbelliferyl butyrate, 2,509 purified designs measured over 2.51 decades). Correlating each metric's *usefulness* on one enzyme against its usefulness on the other asks whether a predictor chosen once can be reused. It cannot.

Rosetta fails differently and more mildly: $\rho_\rho = +0.035$ over 66 shared concepts with 33/66 sign agreement, which is exactly chance. That is *no* transferable signal rather than an inverted one, a weaker claim than AF3's.

The nuance is worth stating, because it is the one place anything survives. PETase's three strongest Rosetta predictors are all His–Asp distance terms, and all three *keep their sign* on `super` while collapsing in magnitude: $-0.502 \to -0.072$, $-0.484 \to -0.023$, $-0.410 \to -0.030$, a 7–21× shrink. Catalytic-geometry strain points the same way on both enzymes and nearly stops mattering; it is the only concept family here that does. Rank agreement over all 66 concepts is still nil, so this does not rescue selection — but it is a different failure from AF3's outright inversion, and the direction is not random.

What *does* replicate is a methodological fact about the two predictor families. On both enzymes, Rosetta carries substantial information AF3 cannot reconstruct while the reverse is largely false: predicting Rosetta features from AF3 gives a median cross-validated R^2 of 0.145 on PETase and 0.199 on `super`, against 0.556 and 0.527 in the other direction. The most private features are relax-ensemble *variance* terms, for which a single structure prediction has no analogue. Same asymmetry, same magnitude, both enzymes.

This is the wet-lab counterpart of the transfer gap in Section 8. There, a policy trained inside a structure predictor scored higher under that predictor than under a held-out one. Here, a predictor that ranks function on the target it was calibrated against anti-ranks it on the next target. Both say the same thing: a structure predictor's agreement with the quantity you care about is a property of the specific system, re-earned per target, never inherited.

That conclusion is an argument *for* the library, not against it. If zero-shot oracle scores could be trusted across targets, the right strategy would be to rank a large virtual set and order the top of it. Because they cannot, the ordering has to be decided by something the oracle does not control — which is what a combinatorially compatible library at a 67% register rate buys: enough physically

Predictor	What it measures	Spearman ρ	n
Zero-shot, AF3			
acyl His107-NE2 pLDDT	catalytic atom, acyl state	+0.60	2,548
acyl active-site pLDDT	active-site shell, acyl state	+0.58	2,548
acyl PAE Ser66↔His107	triad geometry	−0.47	2,548
substrate interface PAE (min)	protein–ligand placement	−0.43	2,616
substrate pTM	global confidence	+0.36	2,616
substrate ipTM	global interface confidence	+0.32	2,616
Zero-shot, sequence only			
mutations from parent	distance, no structure at all	−0.65	3,135
Supervised, held out (RidgeCV, 200 stratified splits)			
AF3 features		$+0.73 \pm 0.02$	
Rosetta features		$+0.75 \pm 0.01$	
AF3 + Rosetta		$+0.80 \pm 0.01$	

Table 1: **Ranking power against measured activity.** Zero-shot rows are univariate Spearman on the screen run. Global confidence is roughly half as informative as a local, state-specific measurement at the catalytic atoms.

assemblable, measurable diversity that the experiment, not the predictor, does the selecting.

5 What a combinatorial library costs

Yours to write. The job of this section, given where it now sits: Section 4 has just established that nothing ranks the candidates, so the obvious move is to order them all. This section is why that is not available, and why the combinatorial form is not merely cheaper but *differently* priced — exponential in k where a one-sequence-per-well library is linear.

The one-line version: a million unique full-length genes is $42M; the same million as a $k = 6$ combinatorial library is under $500.

A protein of L amino acids is $3L$ bases. Cut it into k equal fragments and each fragment is $3L/k$ bases, so a budget B at a price c per base buys

$$v = \frac{B}{3Lc} \quad \text{variants per slot,} \qquad S = v^k = \left(\frac{B}{3Lc}\right)^k \quad \text{recombinants.} \tag{1}$$

There is no k in v: cutting into k pieces buys k times as many fragments, each $1/k$ as long, and the two cancel. Taking logs, $\log S = k(\log B - \log B_1)$ for $B_1 = 3Lc$ the cost of one full-length gene, so on logarithmic axes the model is a fan of straight lines of slope k, all pivoting on B_1.

Numbers that are settled, at Twist eBlock pricing ($0.07 per base, confirmed) for a 200-aa enzyme:

- One full-length gene is $42. A $10 000 order therefore stocks 238 variants in every slot, *however finely it is cut.*
- At $k = 6$ that is 1.8×10^{14} recombinants, against 238 sequences without recombination.
- Reaching 10^{12} recombinants costs $42 000 at $k = 4$ and $1329 at $k = 8$.

And the honest limits, which belong in the text rather than a footnote. The order minimum $m = 300$ bases and adapter overhead $h = 40$ bases are *assumed* and should be checked against a live quote; the price per base is not assumed. Nothing in this section is measured — it is arithmetic on a price — and it says nothing about whether a library *works*, which is Section 6. Two libraries of identical size differ by 50 points of pass rate.

State	Concept	PETase ρ	`super` ρ	Same sign
acyl	His catalytic-N pLDDT	+0.600	−0.147	no
acyl	active-site pLDDT	+0.587	−0.076	no
acyl	His pLDDT	+0.573	−0.080	no
acyl	Ser pLDDT	+0.542	−0.054	no
acyl	Ser-OG pLDDT	+0.526	−0.075	no
acyl	ligand pLDDT (mean)	+0.424	−0.045	no
substrate	Ser pLDDT	+0.385	−0.179	no
delta	interface PAE (min)	+0.371	−0.177	no
substrate	pTM	+0.348	−0.116	no
delta	His catalytic-N pLDDT	+0.345	−0.049	no
mean of these ten		+0.470	−0.100	0/10
all 49 shared concepts		$\rho_\rho = -0.604$		10/49

Table 2: **The zero-shot correlation does not survive an enzyme swap.** Every one of PETase's ten most strongly positive AF3 predictors inverts on a second serine hydrolase. Ranked by $|\rho|$ instead, one of the top ten keeps its sign — the claim is about the predictors anyone would actually pick, which are the positive ones. ρ_ρ is the Spearman between the two enzymes' per-metric Spearmans across all 49 shared concepts: +1 would mean a predictor can be picked once and reused, 0 that one enzyme says nothing about the other, and a negative value that it actively misleads. Sign agreement of 10/49 is worse than the ∼24 expected by chance. Source: `figures/data/petase_super_af3_transfer.csv`.

What binds before the budget does, all three worth stating because they are the reason this section is not the end of the argument: stocking v variants needs v passing designs, so the 256-parent libraries run out at $10 752; one-pot assembly gives $\sim 10^9$ transformants; and the one screen this project has run read 3135 wells. Synthesis sets the ceiling. It has never set the floor — which is what the rest of the paper is about.

6 Results: fragment recombination builds the library

Claim. Designs from the trained policy can be cut at fixed boundaries and recombined across parents, so that $\sum_i |F_i|$ synthesized DNA parts yield $\prod_i |F_i|$ testable variants — and the recombinants keep enough of the parental pass rate that the resulting library is worth building.

Status: scaffold. The computational half is measured; the section is still written as a plan because the blocks below are what the prose has to become. The *physical* half is no longer hypothetical: Section 7 reports a five-fragment library built by exactly this procedure — 128 protein parts spanning 1.1×10^7 recombinants, 3,135 of them assembled and assayed, 67% registering above background at a median of 29 mutations from the parent. That is the one experimental test in this paper of the independence assumption named below.

Why it is the load-bearing section. Section 8 buys a better *distribution*; on its own that is a sampling result. This section is what converts a distribution into something you can physically order: a library assembled from k fragment sets is the product of their sizes, so recombination is what buys library scale that sampling alone cannot. Without it the paper is a benchmark result.

Block 1 — The combinatorial argument, stated precisely

- Set up the notation: L residues split into k fragments at fixed boundaries $[s_i, e_i]$; F_i the set of distinct sequences observed at fragment i across the policy's output; a recombinant is one choice per slot,

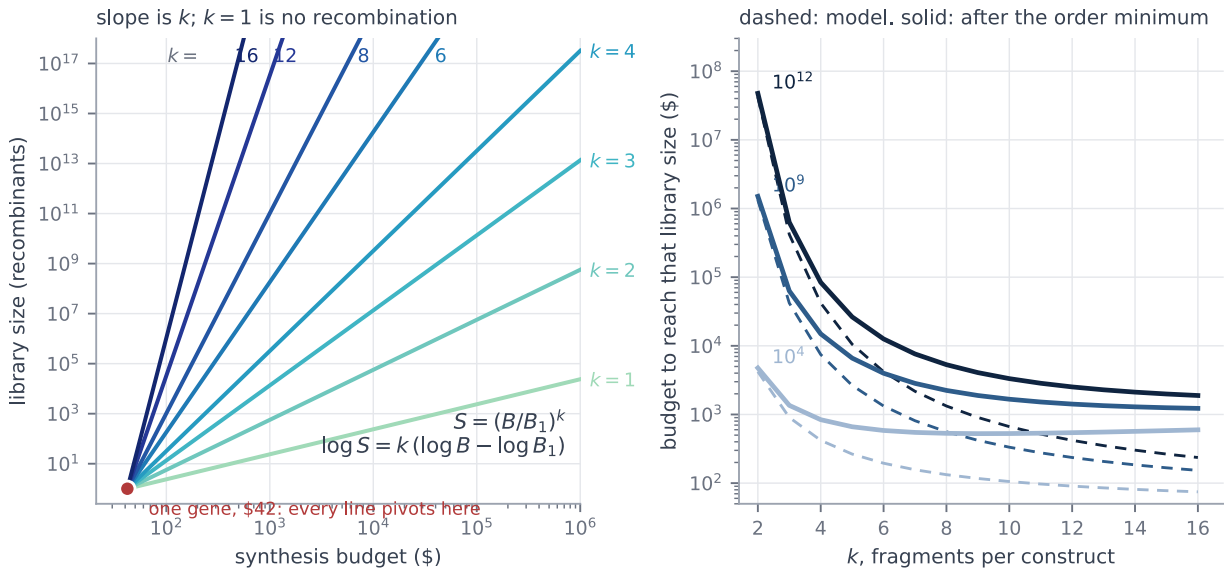


Figure 3: Left: Equation 1 drawn. $k = 1$ is the line without recombination — buy v genes, have v sequences — and the vertical distance from it to any other line is what recombination buys for the same money. Because the axes are logarithmic a different price per base slides the pivot sideways and changes nothing else. Right: the same model inverted — what a target library size costs, and how much of that is the vendor's order minimum rather than the model.

joined.

- The arithmetic: $\sum_i |F_i|$ parts, $\prod_i |F_i|$ recombinants. With $k = 5$ and $|F_i| = 40$ that is 200 parts and 1.0×10^8 variants. **State the actual numbers used, not this example.**
- Golden Gate assembly is what makes the product physically reachable: one pot, orthogonal overhangs at the boundaries, no per-variant synthesis. Methods carries the adapter design.
- **The assumption the whole section rests on, named up front:** fragments are treated as if they contribute independently to foldability. They do not, exactly — the whole point of an inverse folding model is that residues are coupled. Block 3 measures how badly the approximation fails, and that measurement is the result.

[TODO: fragments — state the actual k, $|F_i|$ and library size used for the AME arm. The PETase numbers ($k = 5$, $|F_i| = 25$–26, 1.1×10^7) are in Section 7 and should be quoted here as the physical instance, not substituted for the computational one.]

Block 2 — Where the boundaries go

- Boundaries are chosen on the structure, not the sequence: cut in loops and between secondary-structure elements, never through the active site.
- **Pre-register the comparison:** boundaries placed on secondary-structure breaks should retain more pass rate than boundaries placed at evenly spaced sequence positions. If they do not, the structural rationale is decorative and should be dropped from the paper rather than defended.
- Report the fixed-residue set — catalytic residues held constant across all fragments, so every recombinant retains the active site by construction.

[TODO: fragments — boundary placement rule + the structural-vs-uniform control]

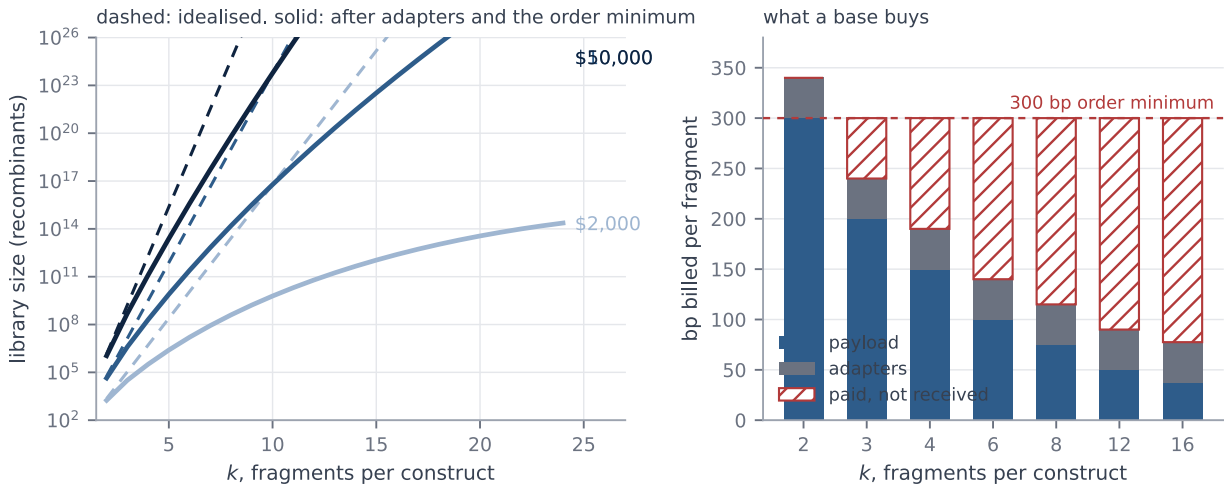


Figure 4: Vendor terms, which Equation 1 ignores. A fragment is billed at $\max(m,\, 3L/k + h)$ for an order minimum m and h bases of Golden Gate sites and overhangs. On a 200-aa enzyme the minimum binds from $k = 3$ onward and by $k = 8$ three quarters of every base bought is padding. The library still grows with k — the exponent beats the division — so cost does not select a k.

Block 3 — Do recombinants fold? (the experiment that must be run)

This is the measurement the section exists for. Everything above is setup and everything below is consequence. Design, in the order it should be run:

1. Sample n designs from a trained CLEO policy with no selection applied, on an AME backbone where the parental pass rate is already known and high (M0097 is the obvious choice — it is the backbone every other measurement in the paper uses).
2. Split each at the fragment boundaries. Keep only fragments drawn from *passing* parents, since that is what one would actually order.
3. Draw recombinants uniformly from the product space, excluding any recombinant that reproduces a parent exactly.
4. Fold best-of-5 and score on the identical code path as Section 8. **Same pipeline, same reference, or the comparison to the parental rate is not a comparison.**

The endpoint is retention: recombinant pass rate divided by parental pass rate. Everything else is secondary.

Pre-registered predictions, written before the run:

- Retention is well above the rate for recombining across *random* MPNN samples at matched diversity. If it is not, fragmentation is adding nothing that temperature would not.
- Retention falls monotonically with the number of slots swapped away from a single parent. A recombinant differing at one slot should behave nearly like its parent; one differing at all k should be worst. A flat curve would mean the fragments are effectively independent, which would be a stronger result than we expect and should be reported as such.
- Recombinants reach mutational combinations absent from the parental set — measured as U over the recombinant library against U over the parents at matched design count.

What kills the claim: retention below the point where the library is filterable in practice. If the parental rate is 57 % and recombinants pass at 2 %, the product space is nominal rather than real, and the honest paper reports the fragment library as a screening funnel with a stated hit rate instead of as a 10^8-member library. Decide the threshold now, not after seeing the number.

This experiment has now been run. The rest of this block reports it. The pre-registration above is left in place so the predictions can be read against the outcome rather than reconstructed after it.

Three AME backbones, 256 designs sampled per arm with no selection applied, split at uniform boundaries into $k = 4$ slots, recombined uniformly over the product space with exact parents excluded, and folded best-of-5 through `af3_from_df` — the identical evaluator used for the parental rate and for every library number in this paper, so the comparison is a comparison.

Table 3: Library evaluation at $n = 256$ parents and $n = 256$ recombinants per cell, $k = 4$ slots, scored by held-out AF3 best-of-5. The two plain-RL rows per backbone are *replicates of one configuration differing only in* `seed`. The bold chimera row is the batch-matched one: $B = 16$, as the plain-RL arms are. The $B = 32$ row is shown because it was the arm reported in earlier drafts, and because the two differ enough to matter. Sample reuse is still unmatched between objectives (4× for chimera, 8× for plain RL); retention is deliberately omitted as a column — see below.

backbone	objective	B	seed	parents	recombinants
M0097	plain RL	16	2	88.7%	68.0%
M0097	plain RL	16	3	44.1%	71.1%
M0097	**chimera**	16	1	84.4%	**85.5%**
M0097	chimera	32	1	94.1%	96.9%
M0255	plain RL	16	2	59.0%	29.3%
M0255	plain RL	16	3	33.6%	30.5%
M0255	**chimera**	16	1	94.9%	**97.3%**
M0255	chimera	32	1	80.5%	82.4%
M0315	plain RL	16	2	89.5%	65.2%
M0315	plain RL	16	3	92.6%	57.0%
M0315	**chimera**	16	1	98.8%	**97.3%**
M0315	chimera	32	1	100.0%	100.0%

Recombinants fold, and training for recombination is what makes them fold. Under the chimera objective at matched batch the recombinant pass rate is *above* the parental rate on M0097 (85.5% against 84.4%) and on M0255 (97.3% against 94.9%), and 1.5 points below it on M0315 (97.3% against 98.8%, where the parents are nearly saturated). Cutting a passing design into four parts and reassembling it across parents costs at most a point and a half of foldability, and on two of three backbones costs nothing at all — so the product space is real rather than nominal.

The comparison that carries weight is against the seed band, not against a single baseline. The two plain-RL rows per backbone are the same configuration run twice, and they bound what a lucky draw can buy. At matched batch the chimera arm clears the full observed band on every backbone: +14.4 points on M0097 (85.5% against 68.0–71.1%), +66.8 points on M0255 (97.3% against 29.3–30.5%), and +32.1 points on M0315 (97.3% against 57.0–65.2%). Measured against each backbone's own seed-to-seed spread in the recombinant rate (3.1, 1.2 and 8.2 points) those margins are 4.6×, 56× and 3.9×.

That is the honest version of a claim earlier drafts made more strongly. The margins quoted previously (+25.8, +52.0, +34.8) came from the $B = 32$ chimera rows, which differ from the plain-RL arms in batch size and sample reuse as well as in objective and oracle. Batch-matching costs M0097 eleven points of margin and gains M0255 fifteen. One caveat survives the correction and should be stated rather than buried: M0097's +14.4 is *below* the 21-point control-pass swing documented between two unseeded runs of a single configuration elsewhere in this work, so on that

backbone the margin alone does not clear the widest variance we have measured. M0255 and M0315 clear it comfortably.

Retention is the least reproducible statistic in this table and should not be read as a point estimate. It is a ratio whose numerator is stable and whose denominator is not. Across the plain-RL replicates the recombinant rate moves by 1.2–8.2 points between seeds while the *parental* rate moves by 3.1–44.5 points, so retention swings from 0.767 to 1.611 on M0097 for a configuration that was not changed at all. It has therefore been dropped as a column: the recombinant pass rate is the endpoint, and it is what the table reports.

Retention is also not the same quantity across arms, which is the second reason to stop tabulating it. The recombination step filters parents to those that already pass, so for an arm whose parents are drawn from a pre-filtered pool the denominator is 1.0 by construction, while for the arms here it is the pass rate of the 256 *sampled* designs. Dividing the unselected-MPNN control by the same rule gives retention of 1.53, 7.59 and 1.43 — values that are obviously not foldability ratios, and that show the column was never comparing like with like. This is a general caution about retention as an endpoint, and it revises the framing of the pre-registration above, which named retention as the endpoint and everything else as secondary.

Scoring the pre-registered predictions. The first holds: recombining across unselected MPNN samples gives a library pass rate of 45.1 % against the chimera arm's 93.1 % pooled over backbones. **The third was ill-posed, and scoring it properly is more informative than the prediction would have been.** It asked whether the recombinant library carries more distinct substitutions than its parents at matched design count. It cannot. Every recombinant is a concatenation of parent fragments, so every (position, residue) pair it carries appears in some parent *at that position*: the recombinant repertoire is a subset of the parental one by construction, and no training objective can change that. Earlier drafts reported the prediction as holding, on a pair of numbers (1,409 against 1,139) that no script on disk produces and that were rarefied at different depths.

Measured as the ratio it actually is, recombination is a mild subset. At matched design count the recombinant library retains 87 % of the parental repertoire at $k = 4$ (0.876, 0.848, 0.880 on M0097, M0255, M0315) and 92 % at $k = 16$ (0.944, 0.913, 0.913), while multiplying the number of variants carrying it by four to thirty orders of magnitude depending on k. That $k = 16$ loses less than $k = 4$, with no overlap between the two ranges, is the expected direction: shorter fragments mean each slot's substitutions recur across more recombinants, so fewer are lost to any particular draw.

The objective does not make the subset gentler — it makes the set being subsetted larger. The preservation ratio is not what the chimera objective improves. Recombining unselected LigandMPNN draws preserves 93 %, marginally the highest of any arm, because a smaller and more redundant parental repertoire has less unique material to lose: that arm retains 836, 584 and 1,222 substitutions out of 854, 668 and 1,294. The chimera arm at $k = 16$ retains 1,095, 1,187 and 1,320 out of 1,160, 1,300 and 1,445 — a slightly smaller fraction of a substantially larger repertoire, and in absolute terms between 1.1 and 2.0 times as many surviving substitutions, at three times the library pass rate. A ratio close to one is cheap to achieve by having little to lose, which is why the absolute count is the quantity that matters here. **The second is not yet measured.** The retention-versus-slots-swapped curve scores recombinants by their Hamming distance in slot space from the nearest parent. That distance is *computable from the scored libraries already on disk* — earlier drafts recorded it as a missing pipeline feature, which was wrong. The real obstacle is the sampling: uniform recombination over the product space at $k = 4$ puts essentially every draw at three slots swapped (249 of 256 on M0097), so the axis has no support at $d = 1, 2, 4$ and the curve cannot be fitted. Measuring it needs a stratified draw at fixed d, not a new metric. It remains the

one pre-registered prediction this experiment does not address, but it is cheap rather than blocked.

The chimera objective buys pass rate by narrowing the policy, and cutting finer buys it back. Pointing the library table at the chimera arms makes a cost visible that a comparison against an obsolete arm had hidden. Against plain RL the chimera arms carry fewer distinct substitutions per library and sit closer together in sequence space, at a markedly higher pass rate:

Table 4: Recombined libraries at $n = 256$, three backbones, AF3 best-of-5, all arms at batch 16. Substitutions are distinct (position, residue) pairs over *passing* members, raw and rarefied to a matched count of passing members. Hamming is the mean pairwise distance within the passing library. `sampled` recombines unselected LigandMPNN draws; it is the no-RL control.

arm	library pass	substitutions (raw)	(matched n)	Hamming
sampled	30.5 %	881	881	75.3
plain RL	76.7 %	1368	1141	76.5
chimera, $k = 4$	93.4 %	1154	895	63.0
chimera, $k = 16$	92.2 %	1201	950	71.2

Three things follow. First, the cost is *within* RL: every chimera arm carries more substitutions than the no-RL control (1,154 and 1,201 against 881) at triple its pass rate, so this is a statement about the objective and not about whether function-aligned search moves the frontier. Second, the cost is concentrated in one cell — M0255 at $k = 4$ falls to Hamming 42.4 against plain RL's 73.2, while M0097 and M0315 sit at 68.6 and 78.0 against 72.0 and 84.4 — and M0255 is also the backbone where the fragment-count sweep is non-monotone, so the two may be one instability. Third, $k = 16$ **dominates** $k = 4$ **on both axes**: more substitutions (950 against 895), greater spread (71.2 against 63.0), the same pass rate, and a product space larger by twenty-five orders of magnitude. The diversity given up by cutting into four parts is largely recovered by cutting into sixteen.

Two limits to state plainly when this is written for submission.

(i) The chimera cells are $n = 1$ in seed. Every chimera-objective run in this paper uses `seed: 1`, and the replicate evidence above comes from plain-RL arms trained against RF3 alone rather than the three-oracle pool. A single-oracle policy can overfit that predictor differently on each run, which is a plausible mechanism for the 44.5-point parental swing and suggests the pool arms may be tighter — but no pool arm has ever been run twice, so the error bar on the chimera row is unmeasured. One replicate of the pool configuration is the cheapest thing that would fix this.

(ii) The contrast varies more than one factor. Batch size is matched in the bold rows, but sample reuse is not: the chimera arms reuse each rollout 4× and the plain-RL arms 8×, because both were configured with `N_updates: 16` at different batch sizes. So the contrast varies objective (chimera vs. plain RL), training oracle (pool vs. RF3 alone) and reuse. It is *consistent with* chimera-plus-pool beating plain-RL-plus-RF3; with one chimera seed and three factors moving it does not establish it, and it does not apportion the gain.

One contrast does isolate the oracle, and it is not the grid this passage used to cite — that grid lives in the project ledger, not in Section 8, and it covers only pool-trained arms, so it cannot speak to pool-versus-single at all. The contrast that can is `fragb32` against `fragb32rf3`: both batch 32 (32 rollouts per step, read off the written metrics rather than the configs), both chimera-objective, both self-resumed with no foreign warm start, differing only in training oracle. Over the held-out AF3 probe points their in-loop overstatement differs by 0.046 (−0.005 for the pool over 14 points against +0.041 for RF3 alone over 10) — so "roughly 0.05" survives, but on this comparison rather than the cited one. It points to the objective carrying most of the effect. Both arms are still $n = 1$ in seed, and this table alone cannot separate the three factors.

Block 4 — Relation to manufacturing-aware generative models

Weinstein et al. [10] is the nearest prior art and the section must engage it directly rather than in passing. Their move is to make the generative model manufacturing-aware from the outset — factorized to match what combinatorial DNA chemistry can physically sample — and they reach $\sim 10^{16}$ designs for antibodies, T-cell antigens and Taq polymerase.

The honest framing. They get there first and at larger scale. What is different is the regime, and it is a real difference rather than a hedge:

- **Their objective comes from a natural distribution.** 300 million observed human antibodies define what a good sample looks like, so the model can be constrained to the chemistry from the start without giving anything up. For a de novo enzyme there is no such distribution — the backbone has no natural homologs — so the objective has to be constructed, which is what Section 8 does.
- **Order of operations.** They factorize first and train the factorized model; we train an unfactorized policy for catalytic geometry and factorize afterwards. That makes retention (Block 3) our problem and not theirs — their samples are manufacturable by construction, ours have to be shown to survive the cut.
- **Scale is not the axis we win on and the section should say so plainly.** 10^{16} is out of reach here and it does not need to be reached: a 10^8 library that is enriched for catalytic geometry is a different object from a 10^{16} library that is not.

Do not write this block as a competitive comparison. The two results compose — a manufacturing-aware architecture trained against a catalytic reward is the obvious next thing, and saying so is more useful than claiming a win.

[TODO: fragments — write Block 4 once Block 3 lands; needs the library-size and retention numbers to be concrete about the 10^8 vs 10^{16} point]

7 Results: a recombined PETase library, assayed

Claim. A fragment-recombination library built by exactly the procedure of Section 6 was ordered, assembled and assayed: 3,135 designs, each 18–76 mutations from its parent, measured on a single run of the MoMI esterase assay. Two things follow. The platform produces physically assemblable, expressible, measurable libraries at combinatorial scale — and zero-shot structure prediction, the signal the policy is trained against, is *lightly* predictive of measured activity on this scaffold — $\rho = 0.58$ at best, worth a 2.8× enrichment, beaten by mutation count alone — and **does not transfer to a second serine hydrolase at all**: across 49 shared AF3 metrics the two enzymes' usefulness rankings anti-correlate at -0.604, and all ten of PETase's most strongly *positive* predictors invert in sign. The oracle calibration is a per-target measurement, not a method property, which is the strongest argument in this paper for deciding a library by experiment rather than by score.

Status: drafted. Every number below is on disk and is regenerated by `paper/figures/petase_zeroshot.py` from `figures/data/petase_activity_zeroshot.csv`; the figure script prints the prose's numbers so the two cannot drift. What this section does *not* contain is a closed-loop activity campaign — see "What this does not show" at the end, and the plan in `archive/results_petase.tex`.

Two subsections moved out of this file on 2026-10-25, verbatim: "Zero-shot prediction is lightly predictive" and "…and it does not transfer to a second enzyme" are now Section 4, where the restart needs them to carry the introduction's ranking claim. The

block above still describes them as part of this section; that framing is yours to redo. What is left here is the physical result — the library was built, assembled and measured.

The library is the method, physically

The assayed library is a five-fragment Golden Gate recombination library over a 209-residue de novo PETase scaffold with the Ser66/His107/Asp122 triad. Boundaries fall at residues giving fragments of 42, 44, 42, 37 and 44 residues. The curated fragment dictionary carries 25, 26, 26, 26 and 25 variants at the five slots: **128 protein parts spanning** 1.1×10^7 **recombinants**, and across library generations 1,920 DNA parts were ordered as BsaI-flanked blocks and assembled one-pot from an acoustic-transfer map, one well per recombinant. This is the arithmetic of Section 6 instantiated in DNA: the ratio of orderable parts to reachable variants is roughly 10^5.

What was recovered and assayed is 3,135 designs carrying a median of 29 mutations from the parent (interquartile range 25–38, full range 18–76). **67% register rate above background**, which is the first result: a library recombined across parents at this mutational depth is mostly not dead. That is the assumption Section 6 names as load-bearing – that fragments contribute roughly independently to foldability – and it is the only direct experimental test of it in this paper.

Searching the library moves activity

Across four design rounds on the single screen run, median activity rises from 2.2 in round 0 to 12.2 in round 3, and the best single recombinant from 20.5 to 31.7 over the same span — with the highest of all four, 47.8, in round 2 (Figure 2b). The progression is not monotone — round 2 has the *lowest* median of the four and the *best* single design, which is what a deliberately broadened library looks like.

A separate, selected set of 20 designs was re-assayed and scored 204–261, roughly 100× the screen median. **Those numbers are from a different run of the same assay** and the two runs share no normalizing control in the record we have, so this paper does not quote a fold-improvement that crosses that line. The 20 also sit 1–11 mutations from the parent, not out in the recombined library, so they are evidence about a local optimization around a good design — not about what recombination reached.

What this does not show

- **No wild-type comparator.** The parent sequence was never assayed in this table, so no fold-over-parent number can be computed from it, and none is quoted.
- **Activity falls with distance.** The most active designs in the record are the closest to the parent. Recombination at 20–80 mutations buys reachable diversity and a 67% register rate; on this scaffold and this assay it did not buy a better enzyme than careful local work did. Section 8's claim is about the diversity/fidelity frontier, and that is the claim this section supports.
- **The oracle calibration does not generalise, and that is measured, not assumed.** A second serine hydrolase inverts every one of PETase's ten best AF3 predictors (Table 2). So this section calibrates the oracle *on this target* and establishes physical realizability; it does not license zero-shot selection on a new target, and no number here should be carried to one. The AME results carry the generality claim.
- **Two enzymes is not a trend.** $\rho_\rho = -0.604$ is one pairwise comparison. It is enough to refuse the claim that the correlation transfers; it is not enough to predict the sign or size of the failure on a third target.
- **The closed loop is not here.** The surrogate-in-the-reward campaign is planned in `archive/results_petase.tex` and has no data in this repository.

8 Results: function-aligned RL moves the diversity/fidelity frontier

Claim. On an established public benchmark, CLEO produces libraries with the sequence coverage of high-temperature LigandMPNN and the pass rate of low-temperature LigandMPNN — a combination no single temperature achieves.

Why this section leads. The PETase result rests on assays that take months and cannot be re-run by a referee. This one is fully computational, uses a published benchmark and a published deposit, and can be reproduced end to end. It buys the method credibility before any wet-lab claim is weighed.

Status: scaffold, and close to **drafted**. Both the CLEO arms and the full temperature ladder are measured. What remains: reconcile the two CLEO rows below (Block 3 quotes the corrected 200-step anchored arm; the notebook's E10 table still carries a superseded pre-fix peak window), rebuild the two PCA panels, fold in the E20 rescue result, and attach the run-to-run error bars from E21.

Block 1 — Setup and the unit of comparison

We evaluate on the atomized motif enforcement (AME) benchmark introduced with RFdiffusion2, comprising 41 catalytic sites scaffolded into 100 backbones each. The published protocol assigns 8 LigandMPNN sequences per backbone and predicts each with 5 Chai models; a prediction succeeds when the motif all-atom RMSD to the design is below 1.5 Å and no backbone atom clashes with the ligand.

The aggregation rule matters more than it looks and is easy to get wrong. We verified it against the deposit rather than assuming: the per-site number reported in `AME_benchmark_summary.csv` is the fraction of the 100 backbones with *at least one* passing prediction out of the 40, matching to zero error across all 41 sites and both methods. It is a best-of-40, not a mean over predictions. Scoring one prediction per sequence and comparing rates therefore understates a comparable method by roughly threefold — in the deposit, only 36 % of the sequences that pass on some Chai model pass on all five. Every number below is on their unit.

- Pilot backbones: three drawn from the `pass` class, at 32/40, 28/40 and 10/40 passing predictions in the deposit.
- Our predictor is AlphaFold3 rather than Chai; best-of-N uses 5 diffusion samples per sequence.
- **Caveat to resolve before submission.** Our implementation of the one predictor-free benchmark quantity, `ligand_dist_des_ncac_min`, does not reproduce the published values: median absolute error 0.4745 Å over 40 backbones, *zero* exact matches (`experiments/ame/calibration_ligand_dist.csv`; the pass/fail call still agrees on 37 of 40). Since it shares parsing and superposition code with the motif RMSD, the absolute values are not yet comparable to theirs. Until it is closed, only the backbone that reproduces (M0097) supports a head-to-head claim.
- **This is a metric-definition gap, not a reference-structure gap.** An earlier draft attributed it to our scoring against the idealized backbone where RFdiffusion2 scores against the unidealized, packed design. That explanation is now excluded by direct measurement. We recovered every candidate design structure from the authors' 177 GB benchmark archive — idealized, unidealized, LigandMPNN-packed, LigandMPNN input backbone — and all four give the same distance to within 0.01 Å while all four miss the published value by ∼0.47 Å. The authors' own `metrics_cache` confirms the mechanism could not have worked: idealization displaces atoms by 0.097 Å mean and 0.274 Å max, an order of magnitude too small. Atom-subset and motif-inclusion variants are likewise excluded.

[TODO: ame — reproduce `ligand_dist_des_ncac_min` exactly; needs `per_sequence_metrics.py`, which is not in the deposit (outputs only). Structure hypothesis falsified 2026-08-15 — do not retry it.]

Scope of this caveat. It bounds only whether our absolute numbers may be set beside the published ones. Every comparison the paper actually makes — CLEO against LigandMPNN, the temperature ladder,

the KL sweep, the objective ablation — runs through a single pipeline with a single reference, and none of them is affected.

Block 2 — Low-temperature sampling returns one solution, sampled repeatedly

The incumbent protocol samples LigandMPNN at low temperature. At $n = 96$ on M0097 this yields a high success rate — 58.3 % of sequences pass — but the passing designs are a tight cluster: mean pairwise Hamming distance 40.6 out of $L = 180$, and 162 distinct substitutions among any 15 of them. The library is close to one solution measured many times, and most of the folding budget spent distinguishing its members buys little.

Raising the temperature removes the redundancy and most of the signal with it. At $T = 1.0$ the same backbone reaches mean pairwise distance 126.3 at a success rate of 4.2 %. Diversity alone is not the objective; a library that cannot be filtered is not a library.

Two numbers here were wrong in earlier drafts and are corrected. $T = 1.0$ was reported as *zero* passing designs; that came from $n = 8$, and at $n = 96$ it passes 4.2 % of the time. And the low-temperature arm was described as collapsing to a single cluster "at every threshold down to 50 % identity" — true at 70 %, but at 90 % every passing design is its own cluster and the statistic simply counts designs. Cluster counts are threshold-dependent and are no longer quoted; mean pairwise Hamming and the distinct-substitution count U are threshold-free and replace them throughout.

Block 3 — cleo occupies the corner neither temperature reaches

Sampling temperature traces a frontier: success falls as diversity rises, and every point on it is available to the incumbent by turning one knob. The claim of this section is that CLEO sits above that frontier, and the measurement is in Table 5.

Table 5: M0097, per-sequence success rate on best-of-5 AF3 predictions, with diversity measured over the *passing* designs only — a library contains only what survives filtering. U_{15} is the number of distinct (position, residue) substitutions in a library of 15 passing designs, rarefied over 300 draws so arms with different yields are comparable. Baselines are $n = 96$; CLEO arms are the final 40 steps of a 200-step run.

Arm	**Success**	**Mean pairwise Hamming**	U_{15}	**Median RMSD** (Å)
LigandMPNN T=0.1	58.3 %	40.6	161	1.42
LigandMPNN T=0.2	66.7 %	47.8	205	1.37
LigandMPNN T=0.3	58.3 %	57.9	273	1.31
LigandMPNN T=0.5	34.4 %	80.5	437	1.70
LigandMPNN T=0.7	26.0 %	100.5	607	1.88
LigandMPNN T=1.0	4.2 %	126.3	—	5.59
CLEO, no anchor	83.1 %	76.6	506	1.07
cleo, KL anchor	56.7 %	**114.4**	**803**	1.44

Interpolating the baseline frontier at the anchored arm's diversity, the incumbent succeeds 14 % of the time where CLEO succeeds 56.7 % — **3.98**×. Read the other way, at CLEO's success rate the incumbent reaches mean pairwise distance 59 against CLEO's 114.4 — **1.93**×. The anchored arm also carries more distinct substitutions per passing design than any baseline ($U_{15} = 803$ against 607 at T=0.7), so this is a genuine exploration advantage and not only a higher yield.

The unanchored arm is the instructive contrast. It reaches the highest success rate in the table, 83.1 %, and the *worst* per-design coverage of any CLEO arm — U_{15} = 506, below T=0.7. Left unconstrained the policy converges onto a narrow solution. The KL weight against the base model is what holds the distribution open, and it is the control the library application needs.

Sweeping that weight over $\{0, 0.005, 0.01, 0.02, 0.05, 0.1\}$ shows it is a well-behaved control with an interior optimum rather than a knob that has to be tuned. Success falls monotonically, 83 % to 10 %, and per-design coverage rises monotonically, U_{15} 506 to 878; the total distinct substitutions a library actually contains is the product of the two and is flat within noise across $w \in [0.01, 0.05]$ (2480 to 2501) before collapsing to 1731 at $w = 0.1$, where too few designs pass for per-design exploration to accumulate. The value used throughout, $w = 0.02$, sits in the middle of that plateau, so the reported numbers do not depend on having found a sharp optimum.

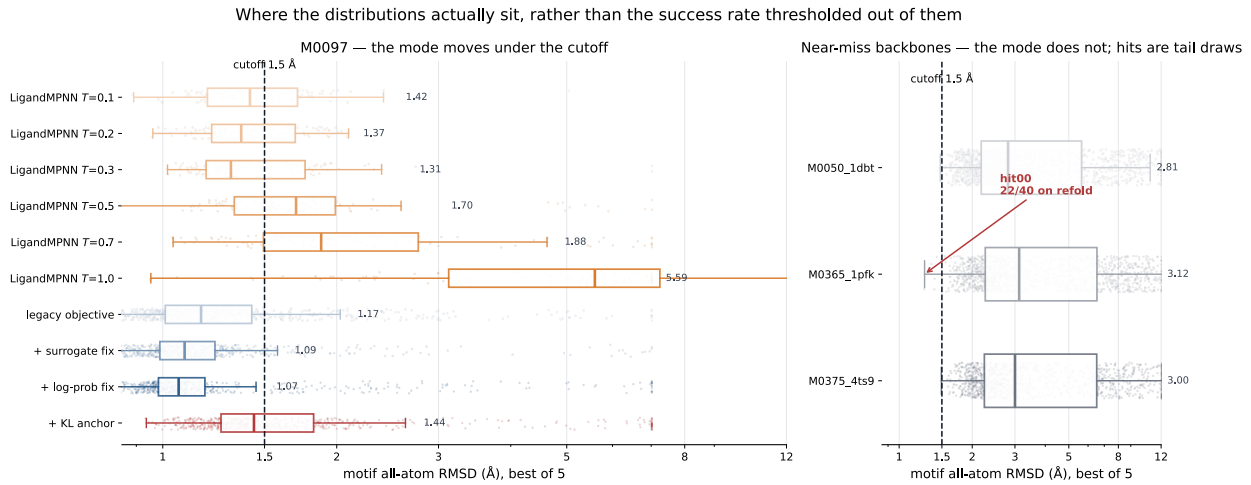


Figure 5: **Where the distributions actually sit, rather than the success rate thresholded out of them.** Each row is one arm's per-design best-of-5 motif RMSD: box for median and quartiles, jittered strip behind it for the shape the box hides. Log axis, because a linear one lets the single worst arm compress everything that matters into the left fifth. **Left, M0097:** every CLEO arm has its *median* below the cutoff (1.07 Å to 1.44 Å) — the mode has moved, not just the tail — against 1.31 Å to 1.42 Å for low-temperature LigandMPNN and 5.59 Å at T=1.0. The non-monotonicity noted in E10 is visible directly: T=0.3 has a *lower* median than T=0.1. **Right, the three near-miss backbones:** medians of 2.81 Å to 3.12 Å with the cutoff far out in the left tail. The one design that survived 40 refolds sits at the extreme edge of its own distribution. Encoding: both ladders are ordinal, so each is a single hue ramped light to dark rather than arbitrary categorical colours; the anchored arm keeps an accent hue as the highlighted result.

Figure 5 shows why the success rate alone is not enough: on M0097 every CLEO arm has its *median* below the 1.5 Å cutoff, so the mode has moved rather than the tail merely reaching across — which is precisely what fails to hold on the near-miss backbones of Block 6.

Block 4 — The training trajectory is the library

This is the reframing that removes a hyperparameter. Rather than adding an explicit diversity term to the reward, we treat every rollout sampled during training as a library candidate. Those sequences were already folded and scored to compute the reward, so the candidate pool is free: the library is a byproduct of training, not an additional expense.

Across 200 GRPO steps at 16 sequences per step, M0097 accumulates 3200 scored designs. Under the anchored policy 1349 of them are distinct passing sequences carrying 3045 distinct substitutions — 96 % of the reachable ceiling. That ceiling is $18 \times 176 = 3168$, not $19L = 3420$: four catalytic positions are held fixed and cysteine is barred from every sampled position, so each of the 176 free positions offers 18 alternatives to its reference rather than 19. The passing set varies at exactly those 176 positions, which is the check that the denominator is the right one.

	cleo, no anchor		cleo, KL anchor	
Cumulative designs	**Unique passing**	**Substitutions**	**Unique passing**	**Substitutions**
400	75	1483	47	1382
800	284	2294	160	2045
1600	893	2805	510	2757
2400	1560	2974	903	2958
3200	2229	3040	1349	3045

The distinct-substitution count is genuinely saturating here, at 89 % to 89 % of the ceiling, while the count of distinct passing sequences is still climbing steeply. The library is not running out of designs; it is running out of *single* substitutions to explore, which is the expected behaviour once most positions have been visited and further diversity must come from combinations.

This supersedes an earlier version of this table reporting 97 passing designs from 3200 folds, a factor of 14 to 23 lower. That run used the pre-audit objective, and its curve flattened because the policy decayed rather than because sequence space was exhausted — the earlier draft said so and predicted rank normalisation would keep it climbing, which is what happened. The pool numbers here are best-of-5, matching Block 3; the superseded ones were single-sample and were not comparable to it.

Block 4b — Where in sequence space the passing designs sit

Generated by `paper/figures/ame_pca.py`, which writes two figures because one projection cannot honestly carry both claims. The shared projection is fit across arms within a backbone — a per-arm PCA places every cloud at the origin and destroys the comparison — but deliberately not across backbones, where a global fit puts 99 % of PC1 variance on backbone identity.

These two panels are stale and must be regenerated. They were built from the pre-audit runs and from baseline arms holding 8 sequences each. Both inputs have been replaced: the objective is fixed and the baselines are now $n = 96$. The drift control below remains the right control and its conclusion is unlikely to change, but no number in these captions should be quoted until they are rebuilt.

Explained variance is low (PC1 5 % to 9 %), as expected for one-hot protein sequence, so the projection is a visual aid; every quantitative diversity claim in this section uses Hamming distance and substitution counts directly.

[TODO: ame — rebuild both PCA panels on the corrected runs at matched n]

Block 5 — The temperature frontier

Measured, and reported in Table 5. Two predictions stated in advance were tested.

The dangerous intermediate temperatures do not close the gap. $T = 0.2$ and $T = 0.3$ were named as the points that would refute a frontier shift if either reached CLEO's coverage at CLEO's success rate. Neither does: both sit below 0.6 of the anchored arm's diversity. The frontier shift stands.

One prediction was wrong. The trade-off was expected to be monotone in temperature; it is not. $T = 0.2$ is the *most* successful arm in the table at 66.7 %, above $T = 0.1$'s 58.3 %, while also

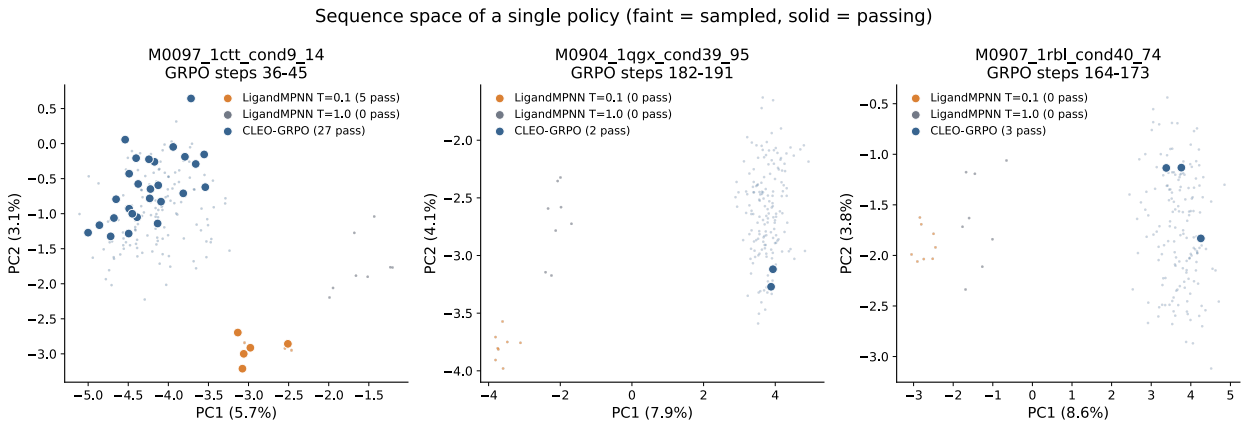


Figure 6: **Sequence space occupied by each design method on the AME pilot backbones.** One PCA per backbone, fit on every sequence that backbone contributes across all three arms, so the arms share one projection and the clouds are directly comparable. Faint points are sampled designs, solid points are designs that pass the benchmark criterion (motif all-atom RMSD < 1.5 Å, no ligand clash). CLEO is restricted to its peak-performing 10-step window so that three *policies* are compared at one point in training. On M0097 the low-temperature arm (orange) forms a tight knot displaced far along PC1 — five passing designs that are a single cluster at every identity threshold down to 50 % — while CLEO's passing designs are spread throughout a broad cloud. T=1.0 (grey) reaches comparable spread with zero passing designs. Projections are deliberately *not* shared across backbones: a global fit places 99 % of PC1 variance on backbone identity, collapsing each backbone to its own blob and hiding the arm differences entirely. Explained variance is low (PC1 5 % to 9 %), as expected for one-hot encoded protein sequence; the projection is a visual aid and every quantitative claim in the text uses Hamming distance and cluster counts directly. Baseline arms hold 8 sequences each, so part of their compactness is small-n.

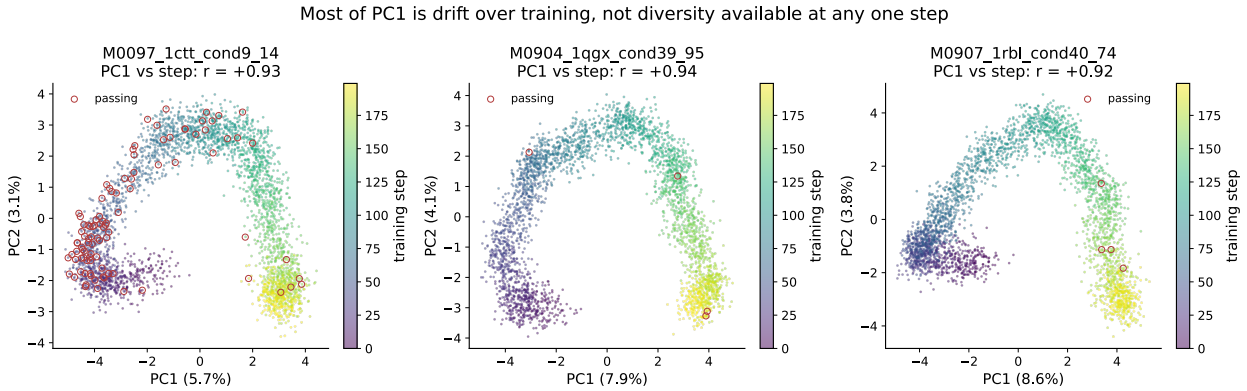


Figure 7: **Most of PC1 is drift over training, not diversity available at any one step.** The full GRPO trajectory for each backbone, in the same projection as Figure 6, coloured by training step; passing designs are circled. PC1 correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones. This distinction is load-bearing rather than cosmetic: spread accumulated by drifting across training is genuine library diversity when the library is assembled from the whole run, which is how ours is assembled, but it is *not* evidence that any single policy is diverse. Presenting the pooled projection alone would conflate the two and overstate the result, so the axis is shown rather than hidden.

being more diverse. $T = 0.2$, not $T = 0.1$, is therefore the incumbent to quote at the low-diversity end, and any comparison against $T = 0.1$ alone flatters us.

Still outstanding from this block's original plan: the saturation fold count for low-temperature LigandMPNN — the point past which it re-emits sequences it has effectively already produced. The CLEO side of that comparison is Block 4; the baseline side needs a cumulative-distinct curve at matched budget, which the $n = 96$ sweep can supply but does not yet.

Block 6 — Rescue of backbones the benchmark cannot solve

Three backbones sit at 1.61 Å to 1.69 Å best-of-40 in the deposit — zero passing predictions, but only ∼0.15 Å from threshold. A pass on any of them is a design the published pipeline cannot produce, with no reproduction confound, because there is no positive result of theirs for us to fail to reproduce.

All three were trained for 75 steps at batch 16. Nine designs cleared the criterion. **Each was then refolded 40 times — the published per-backbone budget — and one survives.** M0365 `hit00` passes 22 of 40 predictions with a *median* RMSD of 1.470 Å, below the cutoff rather than touching it, on a backbone the deposit scores 0/40 with a best of 1.69 Å. That is a design the published pipeline does not produce.

The other eight do not survive. Five pass zero of 40; three pass one or two. Training-time margin predicts this almost perfectly: the single design with a margin above 0.03 Å inside the cutoff is the single design that replicates, and everything at 0.003 Å to 0.023 Å was a tail draw from a distribution whose median sits near 3 Å (Figure 5, right).

The honest claim is one rescue, not nine, and the general lesson is worth more than the rescue: a best-of-N filter applied to a distribution whose mode is far from the cutoff selects excursions, not solutions. A design that truly passes 1 in 40 times is flagged by a best-of-5 filter about 12 % of the time it is folded. Every headline number that leaves this section should be replicated the way `hit00` was, and in-training pass rates should be described as counts of best-of-N events rather than of reproducible designs.

Two caveats remain open. The fold budget is not matched — 1200 designs at best-of-5 against the deposit's 8 sequences at 5 models — so a compute-matched comparison is still owed. And the idealised-versus-unidealised reference structure question from Block 1 applies here as it does everywhere.

Block 7 — The cost axis: evaluations, not steps

Every number in this paper is bought with structure predictions, so the unit that matters is folds, not training steps. The two are not interchangeable: folds per step is rotations × batch × diffusion samples, so at fixed step count a batch-64 run spends eight times what a batch-8 run does. Comparing policies at matched steps compares budgets, not methods.

A batch sweep with sample reuse pinned at 4×, so that batch size is the only factor varying between arms, gives Figure 8. **Batch 16 reaches the** 90 % **chimera pass level on** 40 000 **evaluations against batch 32's** 55 000**.**

The arms are matched in reuse but *not* in total budget, and the difference matters for one claim. Batch 8, 16 and 64 were configured for 48 000, 48 000 and 48 640 folds; batch 32 ran to 105 600. Batch 64 peaked at a smoothed pass of 0.605 and never reached 90 % — but it was never given the 55 040 folds that batch 32 needed to get there, so that is a censored observation and not a measurement. The honest statement is that batch 64 does not reach 90 % within 48 640 evaluations. Batch 8 is cheaper at every level up to 70 % and then needs 46 720 to close, so the ordering inverts between 70 % and 90 % — small batches climb fast and converge late, as expected when the GRPO advantage is normalised by the standard deviation of few samples.

What the in-loop chimera pass rate counts. The training-time chimera set is built by rotation: chimera (ρ, i) takes slot j from parent $(i + \rho j) \bmod B$, so at $\rho = 0$ every slot comes from the same parent and the "chimera" is the unrecombined parent. With four rotations, one in four of the chimeras scored inside the loop is a parent, and the levels quoted above are therefore a quarter parental. This does not distort the comparison between arms — the rotation count is four in every arm, so the parental fraction is identical — but it does mean the absolute level is not a pure recombinant pass rate. Every *library* number in this paper is clean: the library evaluation excludes exact parents explicitly, so the retention table and Section 6 are unaffected.

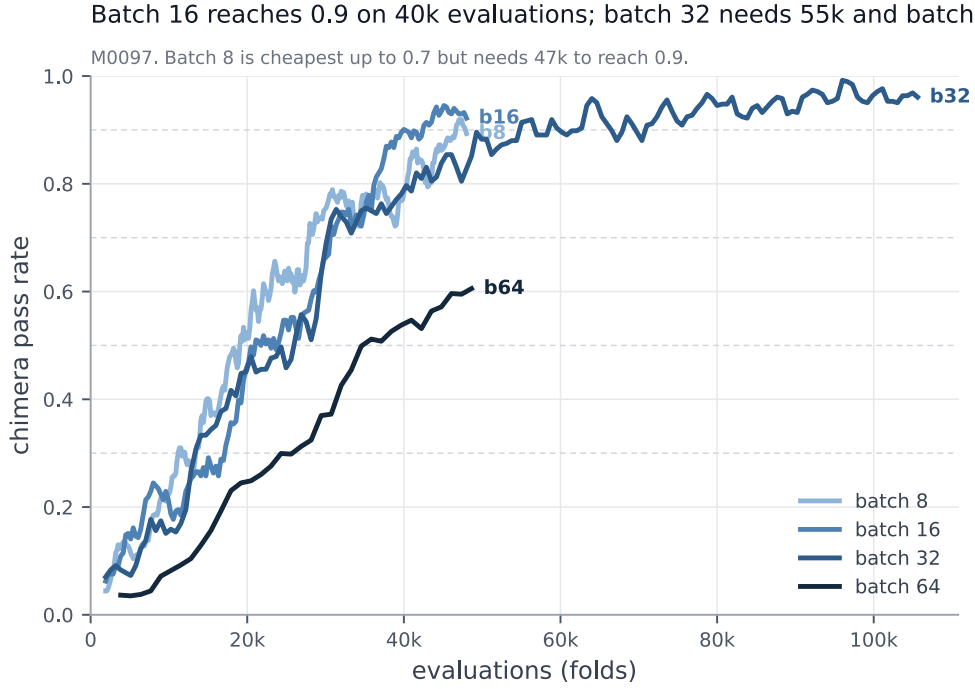


Figure 8: **Evaluation efficiency on M0097.** Chimera pass rate against cumulative structure predictions for four batch sizes at matched sample reuse. Curves are smoothed over a constant *fold* width rather than a constant number of steps: a three-step window spans 480 folds at batch 8 and 1920 at batch 32, which smooths the small-batch arms four times less and lets a noisier curve touch a threshold early by chance. Window width is held at 1920 folds and kept a multiple of three so that the round-robin oracle pool stays balanced.

Why this block is here and not in methods. It reads as a hyperparameter note but it is the reason the held-out AF3 probe grid (project ledger, §4 — not reproduced here) and the library result in Section 6 are affordable at all. It also carries one correction worth keeping visible: every batch comparison made before this sweep held `N_updates` fixed at 16, which makes sample reuse scale as 1/batch — 16× at batch 8 against 4× at batch 32. Those comparisons varied batch and reuse together, so the earlier conclusion that batch 8 has a lower ceiling was not a statement about batch size. Pinning reuse is what makes this one a one-factor result.

Still owed: every cell is $n = 1$ in seed, and the figures above are training-level pass rates. The held-out AF3 probe grid agrees with the ordering (batch 16's final held-out pass on M0255 is 0.938 against batch 32's 0.789), but no arm here has been replicated.

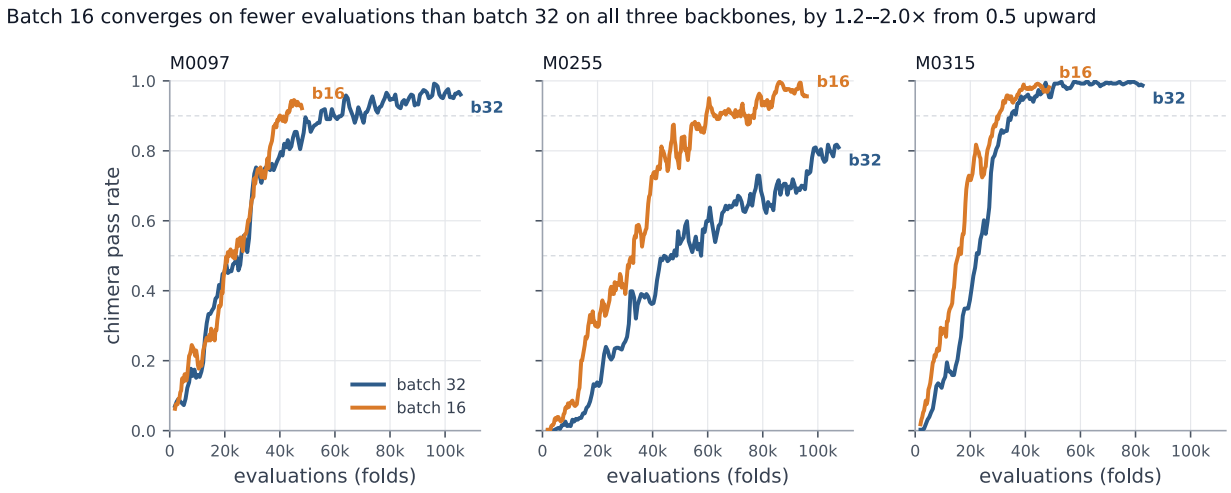


Figure 9: **The ordering replicates across backbones.** Batch 16 against batch 32 on all three AME backbones. The margin is *smallest* on M0097, the backbone the sweep itself was run on. On M0255 — the hardest of the three — batch 16 reaches 90 % at 59 520 folds where batch 32 never reaches it at all within 107 520.

9 Conclusion

Conclusion — yours.

10 Methods: shared pipeline

Shared methodology for the computational results. Per-section specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of an MPNN designer with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [9], ProteinMPNN [1] and LigandMPNN [2].
- **State which base model each result uses.** The AME benchmark arms are LigandMPNN, since those targets have a ligand in the backbone; the PETase campaign is ProteinMPNN. The algorithm is identical across both — only the base policy differs — but a reader cannot be left to infer that from context.

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [11] and diversity-regularized DPO [12]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

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